

Unit

3

Lesson

1



Beyond Imagination.

Tinker Time.

Time Frame: [90 min].

Ready, Set, Learn!

You Need to Know

- Design a 3D printable frame for wearable tech (glasses).
- Use Tinkercad tools to create, align, and group shapes
- Turn 2D sketch or image into 3D using tinkercad.

Challenge questions of the lesson.

1. What design features would make a pair of wearable glasses comfortable and usable for everyday tasks?
2. How can we turn a simple 2D sketch of glasses into a functional 3D model?
3. If someone with a visual impairment were to use these glasses, what design modifications might help improve their experience?

SEL Relation

1. Social Awareness (Empathy and Perspective-Taking):

- Students think about how their glasses design can help real users, especially people who wear them every day or have special needs.
- They consider comfort, ease of use, and how the design could support someone with visual challenges.

2. Responsible Decision-Making:

- Students make smart choices about size, shape, and features to make sure the glasses are usable and safe.
- They fix mistakes, test ideas, and improve their designs by thinking about what works best for the user.

3. Relationship Skills (Collaboration):

- When working in pairs or groups, students share ideas and give helpful feedback.
- They practice listening, explaining their choices, and working together to improve their 3D models.

Engage



Teacher Role:

- Facilitate a class discussion by reading the story aloud with expression.
- Encourage students to imagine themselves as designers solving a real problem.
- Guide the conversation by asking:
 - “What would make smart glasses comfortable or useful?”
 - “Who might need special features in smart glasses?”
- Support students in connecting the dialogue to design features.

Expected From Students:

- Listen to the story and identify design needs mentioned by the characters.
- Think critically about useful features for smart glasses (e.g., adjustable, lightweight, sensor space).
- Write their own ideas clearly and thoughtfully in the space provided.

Parts of the books included.



Story



1. Teacher says (with an enthusiastic tone):

- “Class, you’ve already done an amazing job exploring how smart glasses can help people see and navigate better. But now comes a real designer’s challenge... How can we actually create a frame that holds all the parts and still feels comfortable to wear?”

2. Pause and Prompt:

- “What would you want in a pair of smart glasses to make them both useful and easy to wear?”
 - (Allow a few students to share thoughts about weight, comfort, sensors, etc.)
- Then say:
 - “Great ideas! And in today’s story, Adam and Laila are thinking about the same thing. Let’s see what’s happening in the makerspace...”

Possible Strategies for Implementation:

1. Pre-Reading: Build Curiosity & Context.

- **Activate Prior Knowledge:** Ask, “What do you remember about smart glasses? What makes them different from regular glasses?”
- **Set a Real-World Purpose:** Say, “Engineers don’t just make tech work—they design how it fits into people’s lives. Today, you’ll think like designers too!”
- **During Reading:** Interactive Dialogue Experience
- **Assign Character Roles:** Choose students to read for Mrs. Sara, Laila, and Adam with expression.
- **Use Props or Movement:** Invite students to hold sample frames or gesture where sensors might go as they read their lines.

2. Think-Pair-Share Prompts: After each line, pause and ask students:

- “Why is Laila focused on weight and adjustment?”
- “What do you think Adam means by space for electronics?”

3. Post-Reading: Spark Design Thinking.

- **Discussion Prompt:** “Based on the story, what are the challenges Adam and Laila are trying to solve?”
- **Visual Brainstorming:** Let students sketch what they imagine the smart glasses might look like based on the story cues.
- **Sticky Note Wall:** Students write one “must-have” feature for their frame and post it under categories (comfort, electronics, usability).

Let’s Think!



Objective:

Students will brainstorm essential features for smart glasses by analyzing the story and identifying how form and function meet in wearable tech design.

Answers (may vary):

- Lightweight and comfortable for daily use.
- Adjustable size to fit different users.

- Space to attach sensors and electronic components.
- Durable and safe material.
- Slots for Bluetooth or ultrasonic modules.
- Areas to route wiring without discomfort.

Possible Strategies for Implementation:

1. Pre-Activity

- Ask: “Have you ever worn glasses or sunglasses? What would make them more helpful if they had technology inside?”
- Show simple visual examples of smart glasses or wearable devices.

2. During Activity

- Reread or replay the story dialogue aloud with assigned student roles.
- Pause to highlight key design needs mentioned by the characters (e.g., “adjustable,” “space for electronics”).
- Use a think-pair-share format to allow students to first reflect individually, then discuss with a partner before writing.

3. Post-Activity

- Let students share one feature aloud or draw their smart glasses sketch.
- Organize their ideas into categories (comfort, electronics, design) on the board for later use in the 3D modeling phase.

Explorer's Toolkit



Teacher Role:

- **Facilitator of Design Thinking:** Guide students to visualize and plan their smart glasses layout before jumping into 3D modeling.
- **Ask Guiding Questions:** "Where will the sensor go?" "How will the wires be placed?" "What part should touch the skin or face?"
- **Encourage Creativity:** Prompt students to think about comfort, function, and space when designing their glasses.
- **Monitor Progress:** Circulate the room, check sketches, and offer feedback.

Expected From Students:

- Sketch a labeled diagram of their smart glasses design on paper.
- Show where each component will go (sensor, wires, buzzer, Bluetooth, etc.).
- Consider comfort, usability, and space for wiring in their design.

Parts of the books included.

Explorer's Toolkit



DIY Lab

Grab a pencil and paper and draw your smart glasses design. Plan where each part will go before you start building it in 3D using Tinkercad.



Figure 23: Designing Smart Glasses Prototype.



DIY Lab

Objective:

Students will design a plan for smart glasses by sketching where each component (sensor, buzzer, wires, Bluetooth) will go before building the model in Tinkercad.

Materials Needed:

- Pencil and eraser
- Paper
- Ruler (optional)
- Colored pencils or markers (for clearer labeling).
- Visual of components (can use the illustrated image shown).

Activity Instructions:

1. Look at the smart glasses components.
2. Think about where each piece should go for comfort and effectiveness.
3. Sketch your smart glasses frame and place the components on it.
4. Label each part clearly (e.g., Sensor, Buzzer, Wires, Bluetooth).
5. Use arrows or notes to show how the parts connect.

Possible Strategies for Implementation:

- **Think-Pair-Share:** Let students brainstorm in pairs before drawing.
- **Component Cards:** Give students cutouts or labels of parts to help plan placement.
- **Gallery Walk:** Have students post their sketches and do a peer walk to observe and give feedback.
- **Teacher Modeling:** Model a rough sketch live before letting students begin their own.

Explain



Teacher Role

- **Introduce the Purpose:** Explain that students will learn how to bring their smart glasses design to life in 3D.
- **Provide Support:** Make sure all students can access and view the video.
- **Prompt Thinking:** Ask students to notice how tools like "extrude," "group," or "align" are used in the video.
- **Facilitate Discussion:** Ask students what steps they'll use when they begin modeling.

Expected From Students:

- Watch the video carefully.
- Identify and list key tools used to turn a sketch into a 3D model.
- Be prepared to use these steps in their own Tinkercad design.

Parts of the books included.



Screen Lessons

Click the link to learn how to turn your 2D sketch into a 3D model.

<https://www.youtube.com/watch?v=FkBjRDVel5E>



Screen Lessons

Objective:

Students will understand how to transform a 2D sketch into a 3D model using Tinkercad.

Video Link: <https://www.youtube.com/watch?v=FkBjRDVel5E>



Possible Strategies for Implementation:

1. Pre-Watching:

- Ask: "What parts of your sketch do you think will be easiest or hardest to model in 3D?"
- Set a purpose: "As you watch, notice the steps for building shapes and aligning them."

2. During Watching:

- Pause at key moments and ask: "What tool was just used? Why was it helpful?"
- Students can take quick notes or draw the steps shown.

3. Post-Watching:

- Ask: "What step do you feel most confident about?" "Which part do you think will be tricky?"

- Let students discuss in pairs or jot down steps they will follow for their own 3D build.

Elaborate



Teacher Role:

- Guide students in scanning their sketches and uploading them to the converter tool.
- Support students during the SVG import process in Tinkercad.
- Ask guiding questions to prompt improvement ideas, such as: “Which part of your design seems fragile or exposed?”
- Encourage critical thinking about form, function, and durability in the design process.

Expected From Students:

- Convert their 2D sketch to SVG format and import it into Tinkercad.
- Analyze their digital model and identify areas needing extra protection.
- Justify their reasoning based on potential real-world use (rain, drops, bumps, etc.).
- Begin iterating on their designs based on their analysis and feedback.

Parts of the books included.

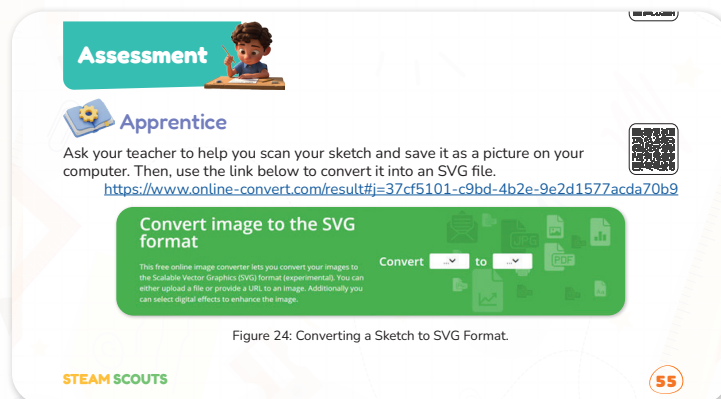


Figure 24: Converting a Sketch to SVG Format.

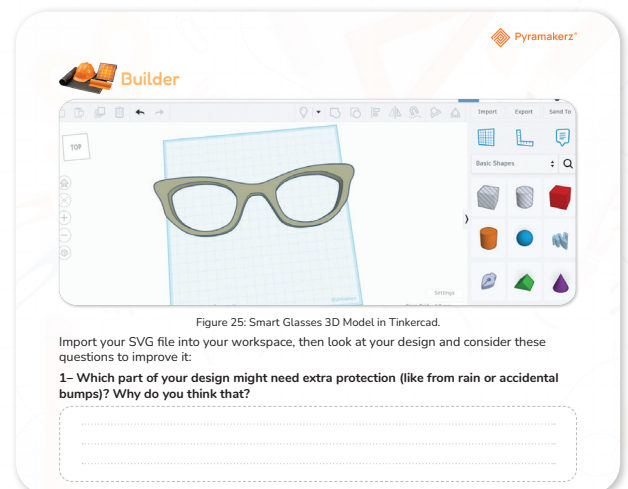


Figure 25: Smart Glasses 3D Model in Tinkercad.

Import your SVG file into your workspace, then look at your design and consider these questions to improve it:

1- Which part of your design might need extra protection (like from rain or accidental bumps)? Why do you think that?



Apprentice

Objective:

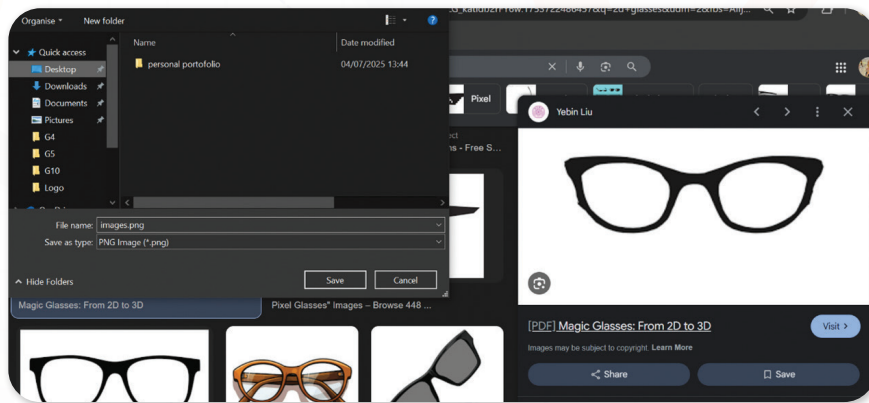
Students will learn how to convert their 2D smart glasses sketches into digital SVG files in preparation for 3D modeling.

Activity link: <https://image.online-convert.com/convert/jpeg-to-svg>

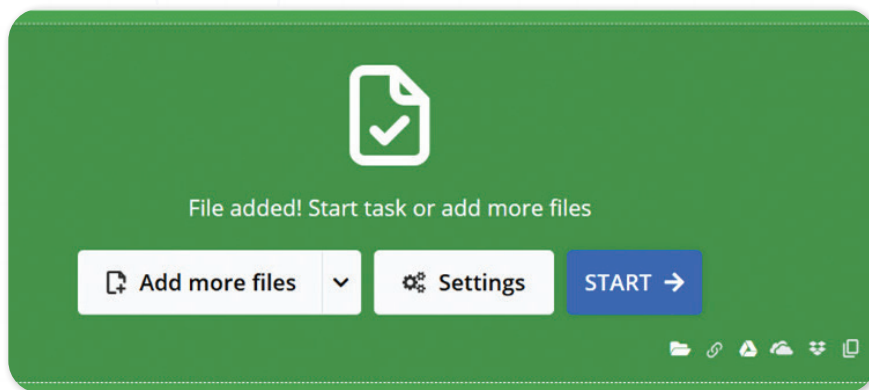
Activity Instructions:

1. Ask your teacher to help you scan or take a clear picture of your sketch.
2. Save the image on your computer or tablet.



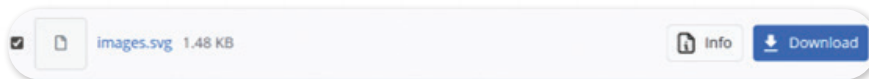


3. Open the provided SVG converter link.



4. Upload your sketch image and convert it into an SVG file.

5. Save the converted SVG for use in Tinkercad.



Possible Strategies for Implementation:

- Demonstrate the conversion process on a projector or shared screen.
- Pair students for peer support—one student handles scanning while the other uploads.
- Prepare a visual step-by-step guide for independent learners.
- Check file compatibility and provide troubleshooting help if the image doesn't convert properly.



Builder

Objective:

- Students will import their SVG file into Tinkercad and critically evaluate their smart glasses design to identify areas that may need additional protection or improvement.

Answer:

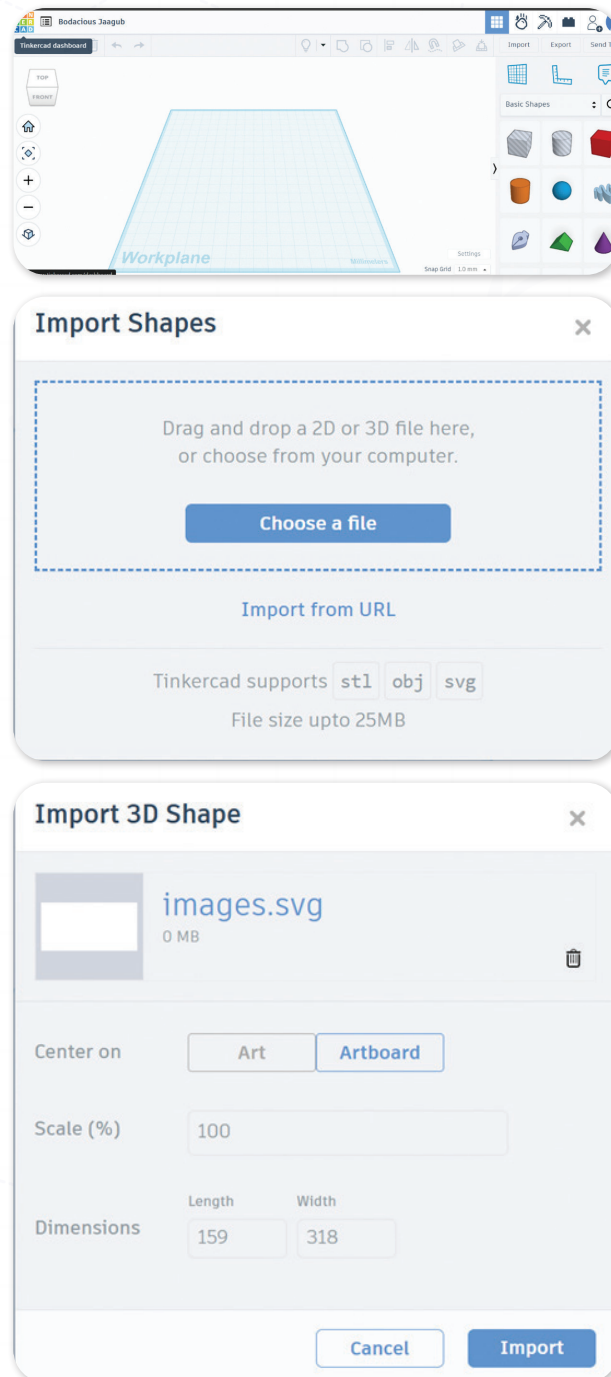
- The front part of the glasses near the sensor might need extra protection because it's more likely to be exposed to rain or get bumped when worn. I think adding a small cover or thicker frame around the sensor area would help keep it safe and working properly.

Materials Needed:

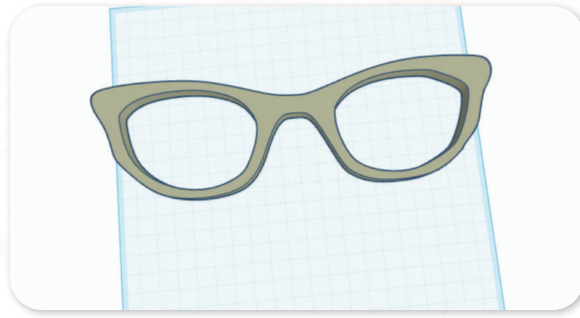
- Computer or tablet with internet access.
- Student's SVG design file.
- Tinkercad account and access.

Activity Instructions:

1. Open Tinkercad and import your saved SVG file into the design workspace.



2. Rotate, zoom, and inspect your model closely from different angles.



3. Think critically about where your design might be vulnerable to damage (e.g., sensor areas, corners, or thin parts).
4. Answer the reflection question:

Evaluate



Teacher Role:

- Guide students in reviewing and refining their designs.
- Provide support in checking space for components (sensors, buzzer, Bluetooth).
- Assist with using Tinkercad tools and help students prepare for 3D printing.

Expected From Students:

- Build the full 3D model of their smart glasses frame.
- Ensure space and comfort features are considered.
- Export the file and ask for printing assistance.

Parts of the books included.



Showcase

Keep working on your smart glasses design by building the frame in Tinkercad. Make sure there's room for all the parts, and try to make the design comfortable and user-friendly. When you're done, ask your teacher to help you print it using the 3D printer.

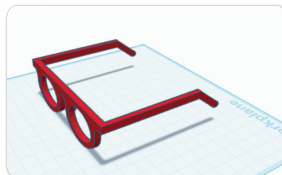


Figure 26: Completed Smart Glasses Frame in Tinkercad.

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Showcase

Objective:

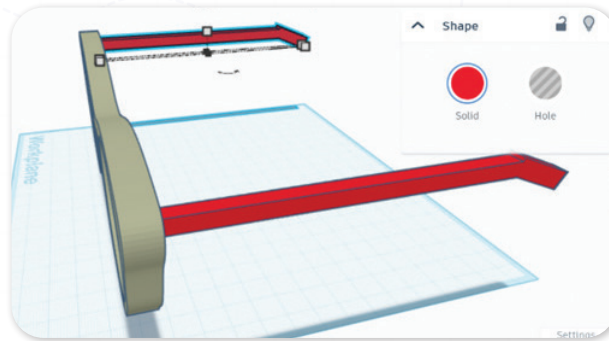
Students will finalize their smart glasses frame design in Tinkercad, ensuring it accommodates all electronic components and is ready for 3D printing.

Answer:

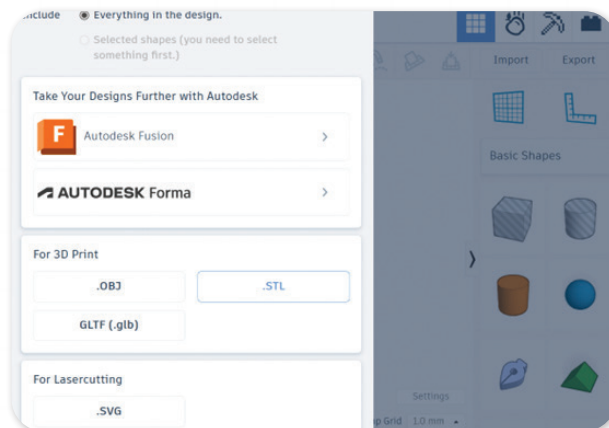
- The design should have room for all components, be adjustable or comfortable to wear, and be strong enough to protect the electronics.

Activity Instructions:

1. Open your Tinkercad project.
2. Continue building and improving the 3D frame.



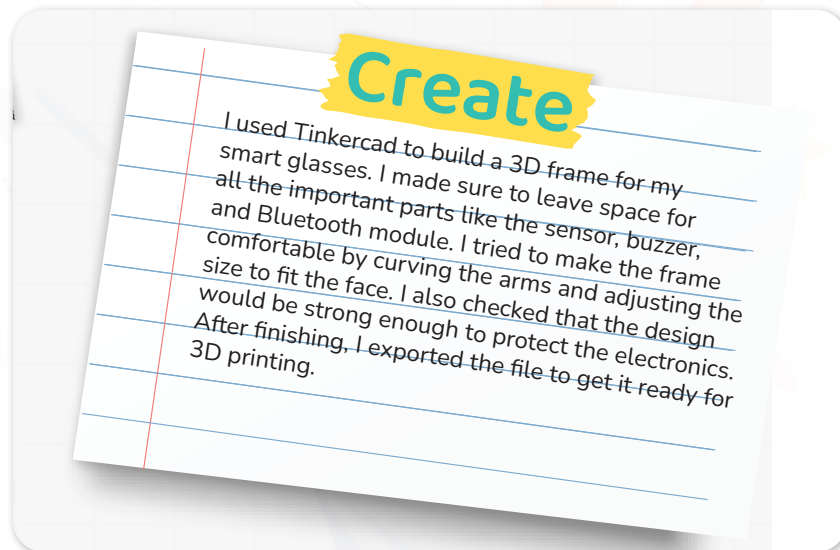
3. Check for room around the eyes, nose, and arms for the components.
4. When finished, export the file and get help from the teacher to 3D print it.



Showcase Rubric:

Criteria	Excellent (4 Point)	Good (2 Points)	Needs Work (1 Point)
Fit for Components	All components fit well with good spacing	Most parts fit; some spacing issues	Missing space for key parts
Comfort and Wearability	Comfortable, well-balanced design	Some comfort issues	Difficult or uncomfortable fit
Creativity and Aesthetics	Unique and appealing design	Basic but neat design	Lacks visual effort or clarity
Tinkercad Skills	Skillful use of align/group/shape tools	Basic tool use	Limited use or frequent errors

It's time to move to the Create part and add this.



Now I Can

Objective:

Students will reflect on their ability to design, model, and transform a 2D idea into a 3D frame for smart glasses using Tinkercad.

Activity Instructions:

1. Introduce the Purpose

- Say: "Let's pause and reflect on what we've learned while designing our smart glasses frame. What skills did you use, and how confident do you feel now?"
- Read Each 'Now I Can' Statement Aloud.
- Read the three statements slowly and clearly.
- After each one, ask students to think: "Did I do this?"

2. Student Response Options

- Students check off the boxes they feel confident about.
- Ask them to circle the one they are most proud of.

3. Optional Sharing

- In pairs or groups, students can share one "Now I Can" statement.
- Prompt: "Which Tinkercad skill helped you the most in building your glasses?"

4. Wrap-Up Reflection

- Ask:
 - "Which skill do you want to practice more?"
 - "How did your 2D sketch help guide your 3D model?"

Relation to Standards.

1. ISTE Standards for Students.

- **4. Innovative Designer:** Students use a variety of technologies (Tinkercad, SVG conversion) to design and test a solution for a real-world problem—creating wearable tech.
- **6. Creative Communicator:** Students clearly express ideas by turning 2D designs into 3D models to communicate their thinking visually.

2. CSTA K–12 Computer Science Standards.

- **2-AP-12:** Design and iteratively develop programs (Tinkercad designs) that include sequences and simple loops to solve a problem.
- **1B-AP-17:** Describe choices made during the iterative design process.

3. Cambridge ICT Starters.

- **Developing Ideas:** Use of 2D/3D digital tools to develop and present original design ideas.
- **Modelling and Simulation:** Creating models (smart glasses) using ICT tools and predicting how design decisions affect the outcome.

4. UNESCO ICT Competency Framework.

- **Knowledge Acquisition & Application:** Learners apply knowledge and technology to solve local, real-life challenges, like assistive technology design.
- **Digital Literacy:** Demonstrates understanding and use of design tools, file formats (SVG), and 3D modeling applications.